

Assignment: Functional Programming in Python

Using `map()`, `filter()`, `sorted()`, and `lambda`

Assignment Objective

The objective of this assignment is to help students understand the **practical and real-world use** of functional programming concepts in Python.

Students will learn how to:

- Transform data using `map()`
- Filter data using `filter()`
- Sort complex data using `sorted()`
- Write clean logic using `lambda` functions

Given Data

```
products = [  
    {"name": "watch", "price": 899},  
    {"name": "laptop", "price": 89999},  
    {"name": "mobile", "price": 9990},  
    {"name": "t-shirt", "price": 999}  
]
```

Instructions to Students

1. Write a **menu-driven Python program**.
2. You must use `map()`, `filter()`, and `sorted()`.
3. You must use `lambda` functions.
4. Do **not** use loops for searching, filtering, or sorting logic.
5. Loops may be used **only for displaying output**.
6. Try solving the problem yourself **before** referring to the solution.

Assignment Tasks

1. Display all product names using `map()`.
2. Search for a product by name using `filter()`.
3. Sort products by price from low to high.

4. Sort products by price from high to low.
5. Filter products whose price is less than or equal to a given value.
6. Exit the program.

Teaching Guide (For Students)

- Use `map()` when you want to **transform** data.
- Use `filter()` when you want to **select** data.
- Use `sorted()` when you want to **reorder** data.
- Use `lambda` to keep your logic short and readable.

Important Reminder: If your solution contains long functions or multiple loops for logic, you are not using functional programming correctly.

Reference Solution

Menu Driven Program

```
products = [
    {"name": "watch", "price": 899},
    {"name": "laptop", "price": 89999},
    {"name": "mobile", "price": 9990},
    {"name": "t-shirt", "price": 999}
]

while True:
    print("\n1. Show product names")
    print("2. Search by name")
    print("3. Price low to high")
    print("4. Price high to low")
    print("5. Filter by max price")
    print("6. Exit")

    choice = int(input("Enter choice: "))

    if choice == 1:
        names = map(lambda p: p["name"], products)
        for name in names:
            print(name)

    elif choice == 2:
        key = input("Enter product name: ").lower()
        result = filter(lambda p: key in p["name"].lower(), products)
```

```
    for p in result:
        print(p)

elif choice == 3:
    for p in sorted(products, key=lambda p: p["price"]):
        print(p)

elif choice == 4:
    for p in sorted(products, key=lambda p: p["price"], reverse=
True):
        print(p)

elif choice == 5:
    limit = int(input("Enter max price: "))
    for p in filter(lambda p: p["price"] <= limit, products):
        print(p)

elif choice == 6:
    print("Program terminated.")
    break
```

Learning Outcomes

After completing this assignment, students will be able to:

- Apply functional programming concepts to real-world problems
- Write cleaner and more expressive Python code
- Understand when to use `map`, `filter`, and `sorted`